# BalanceBoard - A JavaFX Application

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**Abstract**

For as long as education has existed, students have struggled with time management, often overestimating their available time and missing critical academic deadlines. Balancing a busy schedule while knowing when to start each task is challenging, leading to unnecessary setbacks and, for many, an overwhelming cycle of falling behind. However, it doesn't have to be this way. Inspired by my younger sibling's academic hardships in their very first university semester, this application is designed to put an end to time management failures and procrastination, ensuring that completed work gets submitted on time, and further helping students achieve a sustainable work-life balance.

The application provides an intuitive agenda system, allowing users to manage tasks in multiple views, including week, day, and calendar-month modes. Tasks are structured as objects with key attributes like due dates and completion status, enabling efficient organization. To further motivate students, a reward system featuring virtual trophies recognizes task completion and consistency. Additionally, built-in analytics track submission patterns, time taken to start tasks, and other productivity metrics. These insights contribute to a personalized productivity score, providing students with a clear overview of their work habits over daily, weekly, monthly, and yearly periods. Though not initially implemented, AI and machine learning will be used to better cater to each student and their unique work habits, as no two students are exactly the same.

The application is developed using JavaFX for the user interface, offering a responsive and visually intuitive experience for students managing their tasks. The core logic is implemented in Java, utilizing its strong object-oriented paradigm to structure task management, scheduling, and analytics. Data persistence will be handled using SQLite, providing a lightweight yet efficient database solution for storing user tasks, progress, and productivity metrics. In future iterations, machine learning and AI will be integrated to personalize user experiences. This may involve TensorFlow for Java or Weka to analyze study habits and optimize task recommendations based on individual productivity trends. Additional libraries and frameworks, such as JFoenix for enhanced UI components and Quartz Scheduler for task reminders, may also be incorporated to improve functionality and user engagement.

This application is designed for students ranging from high school to university/college, helping them effectively manage their academic workload and deadlines. Recognizing the differences in workflow and academic structure between these levels, a possible feature under consideration is the implementation of two distinct modes: High School Mode and University Mode. This would tailor task organization, reminders, and productivity tracking to better suit the unique challenges faced by students at each stage. While this feature is not yet guaranteed, future iterations may explore its feasibility based on user needs and feedback.

This documentation provides a comprehensive overview of the application, detailing its purpose, design, and technical aspects. It begins with an introduction outlining the app's objectives and the motivation behind its development. The technology stack section describes the key technologies used and their respective roles in building the application. App functionality is then explored, highlighting core features and their significance in enhancing student productivity. Following this, the user interface design section presents the app's visual structure and explains how functionality is balanced with simplicity. The user guide offers detailed instructions on maximizing the application's potential, ensuring users can efficiently navigate and utilize its features. Additionally, data management is discussed, explaining how task-related data is stored, retrieved, and maintained for persistence. Finally, the documentation assesses potential future improvements, including the integration of artificial intelligence and machine learning to further personalize and optimize the user experience.

# 1 Introduction

The transition from high school to university, and even from elementary or middle school to high school, is a significant leap in both academic workload and expectations. Across all levels of education, falling behind in coursework can feel overwhelming, and once a student is struggling to catch up, it becomes all too easy to miss crucial deadlines and face unnecessary penalties.

This issue is personal to me. My younger brother, who had already faced motivational struggles in high school, encountered a particularly difficult first semester in university. Despite having completed most of his work, a simple lapse in due-date awareness led to multiple missed deadlines across all his courses, resulting in a failing CGPA. However, I knew this outcome did not reflect his abilities. Like many students, he simply lost track of deadlines, which is indeed a challenge we all face at some point.

After a deep discussion on motivation and organization, he went on to ace his next semester, proving that success is not a matter of ability, but of structure and awareness. This experience reinforced the importance of maintaining an organized schedule and keeping track of tasks effectively, hence the motivation behind this application.

This application is a comprehensive productivity tool designed to help students effectively manage their academic responsibilities and minimize the impact of procrastination and missed deadlines. At its core, it features an intuitive agenda system that allows users to organize tasks across multiple views, including weekly, daily, and calendar-month modes, catering to different planning preferences. Each task is treated as an object with key attributes, such as a due date and completion status, ensuring structured task management and efficient tracking of upcoming deadlines.

To further engage and motivate students, the app includes a reward system that recognizes consistency and task completion. By earning virtual trophies or achievement badges, students receive positive reinforcement, helping to build better work habits over time. Additionally, the app offers built-in analytics that track key productivity patterns, such as submission trends, time taken to initiate tasks, and overall work consistency. These data points contribute to a personalized productivity score, giving students a clear overview of their performance across various timeframes (daily, weekly, monthly, and yearly).

Future versions of the application will explore the integration of AI-driven insights, which will analyze individual work habits and suggest optimal study schedules based on past behavior. This will allow for personalized recommendations, ensuring that students receive tailored guidance on when and how to structure their workload most effectively. By combining structured task management, motivation-driven rewards, and analytical insights, this application serves as a smart academic companion, empowering students to take control of their productivity and achieve greater success in their studies.

# 2 Research

This section synthesizes existing research on effective study habits and learning environments to inform the design and functionality of the proposed application. Through the combination of educational studies and institutional insights, we aim to identify the behavioral patterns, work habits, and study environments most strongly associated with academic success and translate them into actionable design principles.

## 2.1 Establishing Commonalities in Study Habits Among Successful Students

### 2.1.1 Consulting Sources

It seems appropriate to first establish some common trends between successful students. Beginning with some rather primitive research, this short article from Harvard provides us with a good starting point. It contains 10 major tips and habits to study like a successful Harvard student:

1. "Don't Cram"
2. "Plan Ahead - and Stick To It"

3. "Ask for Help"
4. "Use the Buddy System"
5. "Find Your Learning Style"
6. "Take Breaks"
7. "Cultivate a Productive Space"
8. "Reward Yourself"
9. "Review, Review, Review"
10. "Set Specific Goals"

This is a good place to begin—in fact looking at this list, it is clear that there are already a few areas within the philosophy of this application where we could enhance the development of such habits. For example, "Don't Cram", "Plan Ahead", "Set Specific Goals", "Reward Yourself", and "Find Your Learning Style" are already deeply embedded within the thought and approach behind this app. However, while generally applicable to students of all years of study, it seems more attributed to students transitioning into university from high school. If we want a study that includes all years (1-5) evenly in a challenging program such as medicine, we may choose to consult another source, such as this one from BMC Medical Education. Although it may be so that deep research was present behind the prior source, there is no such documentation. This latter study on the other hand contains a full research paper, complete with numerical results and statistical analysis, using a control group and an exposed group. The main findings can be summarized as follows.

Aljaffer et al. conducted a cross-sectional study among 336 medical students from King Saud University, Saudi Arabia, to investigate the relationship between study habits, personal factors, and academic achievement. The students were divided into two groups based on their Grade Point Average (GPA): high achievers (GPA $\geq$ 4.5) and low achievers (GPA $<$ 4.5). Data were collected through an electronic questionnaire covering demographic information, personal characteristics, and study habits.

The first major type of influence can be categorized as **motivational factors**. Self-fulfillment and ambition were significantly associated with higher GPA scores ($p = 0.04$). Students satisfied with their academic performance were 1.6 times more likely to achieve high GPAs.

The next key type of influence, and our primary focus, are **study habits.**

- Attending lectures and maintaining active attention during classes were linked with stronger academic performance.
- High achievers commonly studied early in the morning, reviewed lecture material beforehand, and maintained weekly study schedules.
- Active recall strategies such as recalling recently memorized material, showed a significant positive correlation with academic achievement ($p = 0.05$).
- Preference for graphical rather than written information was also associated with higher GPA ($p = 0.03$).

Finally, **lifestyle and health factors** such as smoking status, chronic illness, and mental health conditions showed no consistent significant association with GPA, and there is not much our app can do to help with that anyways.

The study concluded that effective study habits and intrinsic motivation are crucial predictors of academic success among medical students. Structuring study routines, employing active recall, and fostering self-driven learning behaviors contribute significantly to higher academic performance. The authors recommend that medical education programs emphasize training in healthy study strategies and consider qualitative research to explore these relationships further.

This source from BMC Medical Education is a very in depth study, and there are many things we can take away from this and factor in the development of the app. For example, when looking at the above motivation factors, we can play into this via a gameified reward system, encouraging messages alongside positive trends (further playing into the fact that students happy with their performance are strongly associated with higher GPAs), etc. Further, when it comes to the main factors, we could play into this via a smart focus mode, limiting app access during class times, monitoring what times they are more focused, and all the other main ideas as discussed in the app's documentation.

Let us look at a third source, such as this one from UC Berkeley.

According to the UC Berkeley Life article *"3 Habits of Successful Students"*, academic success depends less on raw intelligence and more on developing intentional and sustainable daily habits. The article identifies three key habits that consistently differentiate successful students from their peers.

The first habit is **effective time management**. Successful students take control of their schedules by planning ahead and avoiding procrastination. They begin assignments early, maintain organized calendars or planners, and manage tasks using structured blocks of time. Punctuality, consistent attendance, and responding to academic communications promptly are emphasized as integral parts of responsible time management.

The second habit is **regular self-reflection and wellbeing**. Self-reflection allows students to evaluate what study methods and routines are working effectively. This includes prioritizing rest, recognizing signs of burnout, and setting aside time for breaks and self-care. The article stresses that sustainable productivity requires balancing academic demands with physical and mental health, including adequate sleep and mindfulness.

The third habit is **utilizing available resources**. Rather than working in isolation, successful students make use of the institutional and social resources available to them. This includes attending office hours, forming study groups, seeking tutoring or advising, and making use of campus services such as writing centers or wellness programs. Engaging proactively with peers and instructors fosters both academic understanding and community support.

Additionally, Berkeley students interviewed in the article shared several personal strategies:

1. Form meaningful academic and social connections.
2. Complete readings and prepare discussion questions in advance.
3. Find study environments that encourage focus.
4. Set weekly personal goals (e.g., screen-time limits, gratitude journaling).
5. Treat academic work like a structured "9-to-5" routine to protect personal time.

The article concludes that there is no single formula for academic success. Each student must adapt these habits to their personal strengths and circumstances. The overarching principle is consistency: the most effective strategy is the one a student can maintain long-term.

#### 2.1.2 Summarizing the Above Sources

Synthesizing findings from Harvard University, BMC Medical Education, and UC Berkeley reveals a consistent set of study habits shared by academically successful students. Despite differences in methodology and context, these sources converge on several core principles.

1. *Consistent Planning and Time Management.* All sources emphasize the importance of structured planning. Harvard promotes goal-setting and advance planning, Berkeley highlights organization and early task initiation, and the BMC study links weekly study schedules to stronger academic outcomes. Success is strongly associated with maintaining realistic, consistent routines.
2. *Active and Intentional Studying.* Effective learning depends on engagement rather than volume. The BMC study identifies active recall, pre-lecture preparation, and intentional timing of study sessions as strong predictors of performance. This aligns with Harvard's emphasis on repeated review and Berkeley's focus on self-reflective learning strategies.

3. *Self-Awareness and Reflection.* Successful students regularly assess and adapt their study approaches. Berkeley emphasizes reflection and flexibility, while Harvard encourages identifying personal learning styles. Awareness of one's strengths and weaknesses enables more efficient and targeted learning.

4. *Intrinsic Motivation and Positive Reinforcement.* Intrinsic motivation emerges as a key driver of sustained performance. The BMC study found that students satisfied with their academic progress were significantly more likely to achieve higher GPAs. Harvard's recommendation to reward progress reinforces the role of positive feedback in maintaining long-term consistency.

5. *Collaboration and Support Networks.* Both Harvard and Berkeley stress the importance of leveraging external support. Seeking help through peers, tutors, or instructors enhances understanding and reduces isolation, demonstrating that effective learning is often collaborative rather than solitary.

6. *Balance, Wellbeing, and Environment.* Sustainable productivity depends on mental wellbeing and a focused environment. Berkeley highlights self-care and rest, while Harvard encourages breaks to prevent burnout. The BMC findings further associate attentiveness and reduced distraction with stronger academic outcomes.

7. *Overall Pattern.* Across all sources, academic success is best explained not by innate ability but by repeatable behavioral patterns. These findings can be summarized by the following cycle:

   $$\text{Plan} \rightarrow \text{Study Intentionally} \rightarrow \text{Reflect} \rightarrow \text{Stay Motivated} \rightarrow \text{Maintain Balance.}$$

   This cycle forms the behavioral foundation upon which effective learning environments can be built.

## 2.2 Establishing Commonalities in Work Environments Among Successful Students

Although the research discussed above provides adequate detail regarding effective work and study habits, these findings primarily emphasize individual behaviors. It remains to be explored how the broader work and learning *environment* can further accelerate student success. Let us consult this highly detailed research paper from the National Library of Medicine. It provides some truly important insight into this broader topic.

### 2.2.1 The Findings

To complement the behavioral and habit-based findings established earlier, it is equally important to understand the environmental conditions that foster concentration, engagement, and long-term motivation. A valuable study in this regard is that of Rusticus, Pashootan, and Mah (2023), titled *"What Are the Key Elements of a Positive Learning Environment? Perspectives from Students and Faculty"*. The research aimed to identify which aspects of a learning environment contribute most strongly to positive academic and emotional outcomes, drawing upon both student and faculty perspectives.

The authors adopted *Moos' (1973, 1991) model of human environments*, which describes three interacting dimensions that define a positive learning context:

1. *Personal Development (Goal Direction)* - the degree to which the environment supports growth, engagement, and motivation.
2. *Relationships* - the quality of interpersonal interactions between students, faculty, and peers.
3. *Institutional Setting* - the organizational and physical context that provides structure, clarity, and opportunities for community.

Data were collected through five student focus groups and nine faculty interviews at a Canadian university, analyzed using directed content analysis. The study revealed several consistent themes within each dimension:

**Personal Development.** Students reported that engagement and motivation were highest when instructors demonstrated passion for their subject, encouraged interaction, and linked material to real-world applications. Conversely, passive lectures reduced focus and enthusiasm. Faculty echoed this sentiment, noting that their motivation to teach increased when students were attentive and participatory. Both groups also emphasized the importance of managing workload and avoiding burnout—students desired institutional support for stress management and work–life balance, while faculty linked emotional well-being to student responsiveness.

**Relationships.** Interpersonal connection emerged as the strongest determinant of a positive learning environment. Students emphasized that faculty approachability, empathy, and respect created a sense of psychological safety and belonging. Peer collaboration also contributed to engagement, though group work could be frustrating when contributions were uneven. Faculty highlighted similar themes: supportive relationships with colleagues enhanced morale and teaching quality, while positive interactions with students reinforced purpose and fulfillment. However, both groups noted that inconsistent student effort or disengagement could strain these dynamics.

**Institutional Setting.** Both students and faculty identified small class sizes as a key enabler of community, accountability, and individualized feedback. The institution's commuter structure, however, was viewed as a barrier to building social cohesion; limited informal interaction reduced students' sense of belonging. Faculty similarly reported that teaching within a disconnected community diminished engagement and collegiality.

#### 2.2.2 Summarizing the Above Source

In summary, Rusticus et al. (2023) concluded that a positive learning environment is characterized by strong interpersonal relationships, meaningful engagement, emotional well-being, and a supportive institutional structure. The findings validate Moos' three-dimensional framework, emphasizing that relationships are the most influential factor in academic success.

Students and faculty alike valued environments that:

- Promote engagement and intrinsic motivation through interactive, relevant learning experiences.
- Foster psychological safety, respect, and approachability between instructors and learners.
- Encourage community connection through small, cohesive class settings.
- Support emotional well-being and balance to prevent burnout.

These findings align with prior evidence on study habits, reinforcing that learning environments are most effective when they simultaneously encourage structure, reflection, motivation, and a sense of belonging. In the context of this application, such insights can inform the design of digital environments that replicate these qualities—interfaces that feel engaging yet safe, structured yet flexible, and socially connected yet distraction-free.

### 2.3 Building Features and UI's Around the Above Findings

#### 2.3.1 Associating Planned Features with Proven Work and Study Habits

We can construct the following list to draw a connection on what features could enable proven work habits in students:

1. Time management → smart scheduling and anti-cram features
2. Intrinsic Motivation → Motivational messages and gameified reward system
3. Self Awareness → Trend insights, productivity score comprised of various metrics and habit tracking
4. Sustained Productivity → Focus tools and environment optimization such as study-break cycle timer, background noise, etc.

### 2.3.2 Associating Planned UI Implementations with Proven Study Environments

Building on the findings of Rusticus et al. (2023), the design of the application's user interface aims to emulate the qualities of a positive learning environment in digital form. Each visual and interactive component contributes to psychological comfort, focus, and engagement, reflecting the same dimensions identified within Moos' model. While incorporating these principles, the overarching philosophy behind the UI is to remain highly customizable. The core architecture currently supports two interface modes: a *Simple Mode*, emphasizing minimalism and focus, and a *Complex Mode*, offering detailed analytics and interactive components. Beyond these modes, individual menus and elements are fully adjustable, allowing users to tailor their workspace to their personal learning style. The objective is to provide seamless pathways for students to shape an environment that aligns with how *they* learn best.

**1. Psychological Comfort and Focus.** A calm, structured interface mirrors the benefits of a physically organized workspace. The **Simple Mode** embodies this by limiting on-screen elements, using muted color palettes, and presenting information sequentially to reduce cognitive load. These design choices align with research emphasizing the value of minimal sensory distraction for sustained attention and emotional regulation.

**2. Perceived Autonomy and Personalization.** A sense of control over one's environment promotes intrinsic motivation. To reproduce this digitally, the application allows users to customize their dashboard layout, theme, and widget arrangement. This reflects the principle of *perceived autonomy*, a key component of motivation and satisfaction within the learning environment.

**3. Social Connection and Belonging.** As Rusticus et al. highlighted, relationships and community are essential to engagement. The interface incorporates optional collaborative features such as shared study sessions, progress comparison among peers, and gentle accountability reminders. These are designed to simulate the sense of peer connection and support present in an effective classroom environment.

**4. Positive Emotional Climate.** Color temperature, micro-interactions, and subtle animations will be used to convey encouragement and warmth. Positive reinforcement messages, combined with balanced visual feedback (e.g., streaks, achievements), maintain motivation while avoiding anxiety or over-stimulation. This design approach seeks to replicate the positive emotional tone and supportive atmosphere of successful in-person learning environments.

**5. Accessibility and Cognitive Flow.** Finally, consistent spacing, accessible typography, and clearly defined interaction paths create predictability and cognitive ease. This structure parallels the "institutional clarity" dimension of Moos' framework, supporting users through smooth transitions and reliable navigation.

Collectively, these design strategies ensure that the interface functions not merely as a tool, but as a digital environment conducive to focused, self-directed, and emotionally balanced learning. With both behavioral and environmental foundations established, the next step is to translate these design principles into a cohesive system architecture. Each feature and interface element discussed above must ultimately exist as a modular component within the application's object-oriented design. The following section outlines the proposed class structure that underpins the application, beginning with the core `Window` superclass and its specialized child classes (e.g., `WelcomeScreen`, `AgendaScreen`, etc.). This architecture is built around principles of modularity, reusability, and customization—ensuring that the adaptability seen in the user interface is equally reflected in the software's internal design.

### 2.3.3 Tentative Outline of UI and Associated Features

Drawing from the synthesis of behavioral and environmental research, the application's design philosophy translates each pillar of student success into a corresponding set of features and interface components. The resulting framework aims to create a digital environment that both enforces healthy work habits and reflects the structure of an optimal learning space.

**1. Core Environment and User Modes.** The application's dual-interface model provides users with a choice between two main views that primarily affect the welcome screen. Rather than representing rigid configurations, these modes serve as flexible starting points for deeper customization. Users may freely adjust visual density, widget placement, and interface behavior within either mode, ensuring that their workspace evolves alongside their personal study preferences.

- *Simple UI Mode* - This mode aims to reduce noise and provide simplicity. It airs on the side of less noise but more clicks required to access more complex information.
- *Complex UI Mode* - This mode aims to be more immediately informative, but will create more noise and complexity.
- Implementation - The difference between the two will primarily affect the welcome screen, and in the other screens within the application (discussed in detail below) it will result in merely defaulting to their more simple or complex state. But each of these screens will have a toggle to switch them regardless of what UI mode the user chooses.

Together, these modes and their layers of personalization reflect Moos' framework for positive learning environments as interpreted by Rusticus et al. (2023). The freedom to switch between structured simplicity and dynamic interactivity reinforces a sense of *perceived autonomy*, a key contributor to intrinsic motivation and engagement. Furthermore, the customizable nature of the interface encourages users to shape a digital space that aligns with their individual rhythms and learning preferences—reproducing the emotional comfort, clarity, and ownership characteristic of effective physical learning spaces.

**2. Key Feature Mapping.** The major planned features can be directly tied to the research-backed habits and environmental principles discussed above:

- **Smart Scheduling and Anti-Cram System** — Supports time management and consistency through predictive task allocation and reminders.
- **Gamified Reward System** — Reinforces intrinsic motivation by celebrating consistency, focus streaks, and task completion milestones.
- **Focus Tools and Environment Optimization** — Includes features such as a study-break cycle timer, ambient background noise, and limited app access to reduce distractions.
- **Analytics Dashboard** — Promotes self-awareness through trend tracking, performance analytics, and a dynamic productivity score.
- **Collaborative and Social Features** — Encourages accountability and peer support via shared study sessions and progress comparisons.

**3. Emotional and Cognitive Support.** Visual design choices such as soft color palettes, smooth transitions, and positive micro-interactions are informed by research on emotional well-being and engagement. These aim to foster psychological comfort while maintaining cognitive flow, emulating the stability and warmth of positive physical learning environments.

Overall, the application's interface and functionality are not designed merely to organize work, but to actively cultivate the behaviors and environmental conditions proven to enhance learning outcomes.

### 2.3.4 Implementation Details of the Simple and Complex UI Architectures

This section outlines the structural composition of the application's user interface under both the Simple and Complex UI modes, describing the behavior and purpose of each major screen.

**Simple UI Mode**

[*Welcome Screen.*] The Simple UI opens to a minimal landing screen featuring three central tiles—Agenda, Schedule, and Analytics—each providing a brief summary of its respective system. A short *Message of the Day* is displayed to encourage positive engagement. The design prioritizes clarity and low visual noise, allowing users to quickly orient themselves and access core functionality. While additional information is accessible through further interaction, the default presentation remains intentionally sparse.

[*Agenda Screen.*] The Agenda presents tasks in a linear to-do list format, displaying task titles, days remaining, and completion checkboxes. Tasks are automatically ordered by urgency, and recurring tasks may be generated on configurable cycles (e.g., daily or weekly). This structure promotes consistent planning and accountability while maintaining a distraction-free interface.

[*Schedule Screen.*] The Schedule Screen provides a weekly overview of upcoming commitments by default, supporting short-term planning and reducing cognitive overload. Users may optionally switch to broader calendar views. Tasks from the Agenda are automatically integrated, ensuring coherence between planning systems. A toggle allows academic or external events (e.g., classes or meetings) to be displayed without overwhelming the interface.

[*Analytics Screen.*] In Simple Mode, analytics emphasize interpretability over detail. Users are presented with intuitive visual summaries and a single composite **Productivity Score**, broken into three high-level components: *Planning*, *Consistency*, and *Balance*. Weekly trend indicators highlight short-term progress, while lightweight gamified elements (e.g., streaks or achievement badges) provide positive reinforcement. A toggle allows users to transition directly to the more detailed Complex Analytics view.

**Complex UI Mode**

[*Welcome Screen.*] The Complex UI replaces the tile-based landing screen with a denser dashboard layout. Additional elements—such as expanded analytics previews, subdivided scheduler components, and auxiliary tools—are visible immediately. Despite increased information density, components remain visually grouped and structured to preserve navigability.

[*Agenda Screen.*] The Agenda Screen remains unchanged between UI modes. Its task-focused nature necessitates a consistent, minimal design to avoid unnecessary cognitive load.

[*Schedule Screen.*] The Complex Schedule defaults to a monthly overview, encouraging longer-term planning while retaining toggle options for weekly and calendar-style views. Academic events and task deadlines may be viewed simultaneously to support holistic workload management.

[*Analytics Screen.*] The Complex Analytics Screen exposes the full analytical framework underlying the Productivity Score. Each contributing metric—such as task initiation timing, completion consistency, and workload distribution—is displayed numerically alongside trend indicators. Interactive filters and hover-based explanations allow users to explore behavioral patterns and temporal trends in greater detail. Gamified achievements adapt dynamically to measurable progress, reinforcing motivation through transparent feedback.

# 3 Technology Stack

# 4 App Functionality

## 4.1 The Core Feature

The core functionality of this application revolves around the agenda system, an enhanced to-do list that allows students to efficiently track assignments, deadlines, and study plans. Tasks are displayed along with their respective due dates and time remaining, helping students visualize upcoming responsibilities. Users can add, remove, and complete tasks, with completed tasks disappearing in a 'successful completion' manner. Additionally, tasks can be set as recurring, which is ideal for assignments like weekly quizzes or discussion posts, ensuring that repeated deadlines are never overlooked.

The agenda will provide multiple viewing options to accommodate different planning styles. Users can switch between a daily view, a weekly view, and a calendar-style month view. To avoid overwhelming the interface, the monthly calendar view will display only hard deadlines and release dates, keeping the display clean and focused.

## 4.2 Analytics and Productivity Tracking

Beyond basic task management, the application continuously runs analytics in the background to provide meaningful insights into a student's workflow. While more analytics may be added in the future, the following key metrics will be included in the initial implementation:

- Time elapsed between the release date of a task and when the student starts working on it.
- Time taken between task completion and final submission.
- Weekly and monthly distribution of workload - analyzing whether work is balanced daily or left to accumulate before deadlines.

These insights will contribute to a personalized productivity score, providing students with a clear and measurable depiction of their work habits over weekly, monthly, and potentially even yearly periods. This score aims to encourage self-improvement by helping students recognize tendencies such as procrastination or last-minute cramming.

## 4.3 Task Prioritization and Smart Scheduling

A dynamic task prioritization system will assist students in structuring their workflow more effectively. By considering assignment weight and due dates, the system will generate a personalized study plan that ensures high-impact assignments receive appropriate attention. Rather than simply listing tasks in chronological order, this feature helps students allocate time based on importance, preventing last-minute rushes and optimizing productivity.

## 4.4 Gamified Reward System

To further motivate students, a reward system featuring virtual trophies and achievements will recognize task completion and consistency. By gamifying the study process, this system fosters positive reinforcement, keeping students engaged and on track.

## 4.5 Built-In Grade Calculator

A feature that could prove to be very useful is a built - in grade calculator system, that is automated, pending the user's input grades and grading schemes. I personally have a routine that I execute come finals season where I make a list of all attainable final grades in the class and the corresponding required final exam score, giving me a foolproof look at how I need to perform - it is useful in cases where you need to simply pass, and also useful in cases where you need that A. In any case, it allows for strategic studying rather than relying on guesswork of how you need to perform, which has surely caused some preventable failures in the past.

The built-in grade calculator provides a seamless way for students to track their academic standing and determine required scores for their desired final grade. While not mandatory, this feature is available for users who wish to take advantage of it.

Students can customize their grading setup by adding exams, assignments, quizzes, papers, and other coursework, each with its respective weight. The system also supports grading rule variations, such as:

- **Dropped grades -** many courses offer schemes such that the lowest assignment, quiz, or other item is dropped.
- **Multiple grading schemes -** most courses offer multiple schemes and accept the highest grade.

Behind the scenes, the grade calculator is structured using object-oriented programming (OOP) principles. Each coursework entry (exam, assignment, etc.) is treated as an object, storing its weight and grade as fields. These objects are then processed through a calculation array, efficiently computing weighted averages and final grade projections.

This approach promises that the system remains scalable, efficient, and easy to update, providing students with a clear, stress-free way to stay on top of their grades.

### 4.6 Future AI/ML Enhancements

While the initial version of this application focuses on structured task management and analytics, future iterations will explore AI and machine learning to further optimize student productivity. Potential AI-driven enhancements include:

- **Personalized task suggestions** – Recommends optimal start times based on known work habits.
- **Smart reminders** – Adaptive notifications that trigger at moments when procrastination is most likely.
- **Predictive analytics** – Forecasts productivity trends and suggests improvements.

These features would allow the application to adapt to individual student behaviors, making it an even more powerful tool for academic success.

## 5 User Interface Design

### 5.1 Launching the app for the first time (Setup Screen)

What do we need to know about the user? Certainly not very much - there is no personal information required. We merely seek academic details, such as the number of courses, number of credits per course, grading scheme When a new user launches the application for the first time, they will be prompted to answer a few short questions, including:

1. Are you a university student, or a high school student?
   - If university student:
     - What is your program, year, and school?
     - to begin a new term within this system, please enter your courses for this semester along with their credit amount.
   - If High School Student:
     - What grade?
     - what country/state/province (Your school's location helps determine your school's grading scale and term structure.)
     - what style of term system do you have? (e.g. yearly, two semesters a year, quarter system, etc)
     - to begin a new term within this system, please enter your courses.
2. Do you prefer having everything listed at once (i.e. can see everything at a glance even if it means more clutter), or do you prefer a simple UI (i.e. few things on the screen at a time, less overwhelming but requires extra navigation to find certain info)? (This is to tailor the UI to suit their learning style)
3. Do you prefer a system that actively reminds you of tasks, (e.g. push notifications, reminders when procrastination is detected) or a system that simply lists things (no interruptions, "do not disturb" - just a clean space you check on your own)?
4. (optional) Are there any issues or challenges that have historically affected your grades or learning (e.g., procrastination, burnout, test anxiety, ADHD, etc.)? We'll use this info to better tailor reminders, notifications, and UI interactions to support your unique needs - no judgment, just support. You can update this anytime. We want BalanceBoard to grow with you, not just around your deadlines.

**Note:** Everything here can be changed later in Settings. Your answers to this questionnaire are always saved — you can revisit and edit them anytime.

In Settings, users can revisit the original onboarding questionnaire. Each answer appears alongside the unselected options, allowing users to quickly remember their previous choice and explore alternatives. Tooltips above unselected options provide a short summary of what would change if the option were selected — making experimentation frictionless and fully reversible.

## 5.2 Launching the app regularly (Welcome Screen)

The regular user interface will be different depending on how the user answered question (2) in section 4.1. As a general rule, simplicity should always be of primary importance, but this simplicity will appropriately be balanced with detail and complexity as per the users wishes and can be changed at any time. It should be noted that in each version, the set of features are identical, the choice merely affects the amount and type of information *immediately available.*

### 5.2.1 The User Prefers Simplicity

Upon launching the app, the UI takes a minimalist approach. There will be three windows (stages) visible in a row on the screen to start, along with a large welcome text above, that changes slightly each time the app opens. The three clickable windows, in a row from left to right, are as follows:

- Todo list, with a select few of the most important tasks visible through the window before clicking. After launching this window, you gain access to the full list where you can check off tasks as well as add or remove tasks, and other general modification.
- Schedule, initially through the window you'll see the most imminent upcoming deadlines, and upon clicking launches into a weekly teams-style schedule, and you have the option to zoom out into a calendar view, where you can scroll between months
- Analytics window - a quick view of your work habit trends and where you can improve as well as where you excel. Upon launching this window, you'll gain the fully detailed view, complete with all your start/submission logs, grades, etc.

### 5.2.2 The User Prefers More Detail/Complexity

Here, upon launching the app, the user will be taken to more of a dashboard-style view, with the most information available immediately. For now, the idea for the dashboard screen to be in a tiles style, though this may take other forms later down the line as I, and the users see fit. The main goal would be to be able to access every aspect of the app and all features through one click (whereas in the simpler version, the tradeoff for less 'clutter' is that you may have to click more than once to get to where you want to go in the end). Both styles will try to balance these aspect, they are merely tuned closer to opposite ends of that scale.

# 6 User Guide

# 7 Data Management

# 8 Future Improvements & AI Integration

# 9 Conclusion